A FOURTH MOMENT ESTIMATE FOR LOGARITHMIC INTEGRALS OF RANDOM LITTLEWOOD POLYNOMIALS AND ERDŐS'S ROOT-COUNT PROBLEM

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Abstract. Let $P_n(z) = \sum_{k=0}^{n-1} \varepsilon_k z^k$ be a random Littlewood polynomial and let

$$L_n(r) = \int_{\mathbb{T}} \log \left| \frac{P_n(re^{i\theta})}{\sigma_n(r)} \right| d\mu(\theta), \qquad \sigma_n(r)^2 = \sum_{k=0}^{n-1} r^{2k}.$$

We prove a fourth-moment concentration estimate, uniform in the thin annulus $r \in [1 - n^{-11/10}, 1 + n^{-11/10}]$: there are absolute constants $C_{\mathrm{KL}}, \kappa_{\mathrm{KL}} < \infty$ such that

$$\|L_n(r) - \mathbb{E}L_n(r)\|_{L^4(\Omega)} \le C_{\mathrm{KL}} (\log n)^{\kappa_{\mathrm{KL}}} n^{-1/2}.$$

Together with Yakir's first-moment estimate, this gives the corresponding uncentered fourth-moment bound for $L_n(r) + \gamma/2$. The proof combines a fixed-dimensional anti-concentration estimate for Rademacher sums, occupation bounds for the small values of $P_n(re^{i\theta})$, a logarithmic smoothing at scale $(\log n)^B n^{-1/2}$, Pisier's inequality on the discrete cube, and the logarithmic integrability theorem of Nazarov–Nishry–Sodin. As an application, Jensen's formula converts the logarithmic estimate at three nearby radii into a quantitative almost-sure root count. For the Erdős formulation, if R_n denotes the number of zeros of P_{n+1} in $\{|z| \le 1\}$, counted with multiplicity, then almost surely, for every $\delta > 0$,

$$R_n = \frac{n}{2} + O_\delta(n^{7/8+\delta}).$$

In particular, $R_n/(n/2) \to 1$ almost surely.

1. Introduction

Erdős asked whether a random Littlewood polynomial

$$Q_n(z) = \sum_{k=0}^{n} \varepsilon_k z^k, \qquad \varepsilon_k \in \{-1, +1\} \text{ independent and symmetric,}$$

has asymptotically half of its zeros in the closed unit disk, almost surely [1, p. 252]. In other words, if R_n counts the zeros of Q_n in $\{|z| \le 1\}$, with multiplicity, the question is whether

$$\frac{R_n}{n/2} \longrightarrow 1 \qquad \text{almost surely.}$$

Yakir proved the corresponding convergence in probability, with a quantitative form showing that deviations of order $n^{9/10}$ have probability tending to zero [4]. The purpose of this paper is to upgrade the available probabilistic control to a summable fourth-moment estimate and thereby obtain an almost-sure result with a polynomial error term.

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We work mostly with the notationally convenient polynomial

$$P_n(z) = \sum_{k=0}^{n-1} \varepsilon_k z^k,$$

which has n coefficients. Passing from P_n to Erdős's Q_n amounts only to replacing n by $n+1$. For $r > 0$, set

$$\sigma_n(r)^2 = \sum_{k=0}^{n-1} r^{2k}, \qquad Z_r(\theta) = \frac{P_n(re^{i\theta})}{\sigma_n(r)},$$

and consider the normalized logarithmic average

$$L_n(r) = \int_{\mathbb{T}} \log |Z_r(\theta)| \, d\mu(\theta).$$

The normalization removes the deterministic radial growth of P_n. Yakir's first-moment estimate gives

$$\mathbb{E}L_n(r) = -\frac{\gamma}{2} + O\left(\frac{(\log n)^2}{\sqrt{n}}\right)$$

uniformly for r in the annulus $[1 - n^{-11/10}, 1 + n^{-11/10}]$. Thus the main probabilistic task is to show that $L_n(r)$ is concentrated around its mean on the same $n^{-1/2}$, up to polylogarithmic losses, scale.

Our main analytic estimate is the centered fourth-moment bound

$$\|L_n(r) - \mathbb{E}L_n(r)\|_{L^4(\Omega)} \le C_{\mathrm{KL}} (\log n)^{\kappa_{\mathrm{KL}}} n^{-1/2}, \qquad r \in [1 - n^{-11/10}, 1 + n^{-11/10}],$$

for absolute constants $C_{\mathrm{KL}}, \kappa_{\mathrm{KL}} < \infty$. Combined with the first-moment input, this yields

$$\mathbb{E}\left|L_n(r) + \frac{\gamma}{2}\right|^4 \le C(\log n)^{\kappa} n^{-2}$$

uniformly throughout the same annulus. This estimate is the quantitative replacement for independence that is needed in the Borel–Cantelli step: it allows us to control $L_n(r)$ simultaneously at the three Jensen radii $1 - \tau_n, 1, 1 + \tau_n$, where $\tau_n = n^{-9/8}(\log n)^{\kappa_{\mathrm{root}}}$.

The proof of the fourth-moment estimate has three main ingredients. First, we prove occupation estimates for the small values of $f_r(\theta) = P_n(re^{i\theta})$. If

$$M_{n,r}(u) = \mu\{\theta \in \mathbb{T} : |f_r(\theta)| \le u\},$$

then, for $1 \le p \le 6$ and $0 \le u \lesssim \sqrt{n}$, we show

$$\|M_{n,r}(u)\|_{L^p(\Omega)} \le C_p \frac{(1+u)^2}{n}.$$

The estimate is obtained by decomposing p-tuples of angles into clusters under the symmetries $\theta \sim -\theta$ and the two exceptional poles $0, \pi$. For well-separated representatives, a mixed covariance lower bound reduces the small-ball estimate to a fixed-dimensional anti-concentration lemma for Rademacher sums. Components near 0 or π contribute one real constraint; non-pole components contribute two real constraints.

Second, we replace $L_n(r)$ by the smoothed logarithmic functional

$$F_\alpha = \frac{1}{2} \int_{\mathbb{T}} \log\big(|Z_r(\theta)|^2 + \alpha^2\big) \, d\mu(\theta), \qquad \alpha = (\log n)^B n^{-1/2}.$$

The occupation bounds imply moment bounds for the resolvent

$$R_\alpha = \int_{\mathbb{T}} \frac{d\mu(\theta)}{|Z_r(\theta)|^2 + \alpha^2}.$$

A continuous gradient estimate for F_α, together with a finite-difference comparison along sign flips, bounds the discrete carré-du-champ of F_α by $n^{-1}(R_\alpha + R_\alpha^2)$. Pisier's inequality on the discrete cube then gives fourth-moment concentration for F_α.

Third, we remove the smoothing. The difference $F_\alpha - L_n(r)$ is bounded by a resolvent term and by an integral of the occupation function over small levels. The midrange levels are controlled by the occupation estimates, while the deepest levels are handled using the logarithmic integrability theorem of Nazarov–Nishry–Sodin. This gives

$$\|F_\alpha - L_n(r)\|_{L^4(\Omega)} \le C(\log n)^\kappa n^{-1/2},$$

which transfers the concentration estimate from F_α back to $L_n(r)$.

Finally, Jensen's formula turns the logarithmic estimates into zero counts. The deterministic part $\log \sigma_n(1 \pm \tau) - \log \sigma_n(1)$ contributes $n/2$ up to an error $O(1 + \tau n^2)$. On the almost-sure event supplied by Borel–Cantelli, the random logarithmic increments contribute $O(b_n/\tau_n)$, where $b_n = n^{-1/4}(\log n)^{2\kappa_{\mathrm{root}}}$. The choice of τ_n balances these two errors and gives

$$R_n^< = \frac{n}{2} + O\big(n^{7/8}(\log n)^{\kappa_{\mathrm{root}}}\big), \qquad R_n^\le = \frac{n}{2} + O\big(n^{7/8}(\log n)^{\kappa_{\mathrm{root}}}\big)$$

almost surely for the n-coefficient model. Applying this to P_{n+1} yields the announced Erdős formulation.

The main contributions of the paper are therefore:

(i) a uniform fourth-moment concentration estimate for logarithmic integrals of normalized random Littlewood polynomials in an $n^{-11/10}$-neighborhood of the unit circle;
(ii) a self-contained occupation-measure argument based on finite-dimensional small-ball estimates and cluster bookkeeping near the exceptional points 0 and π;
(iii) a smoothing-and-removal scheme that combines resolvent bounds, a finite-difference carré-du-champ estimate, Pisier's inequality, and Nazarov–Nishry–Sodin logarithmic integrability;
(iv) a quantitative almost-sure version of Erdős's root-count problem: $R_n = n/2 + O_\delta(n^{7/8+\delta})$ for the closed unit disk.

2. Statement and cited estimates

Let

$$\Omega := \{-1, +1\}^{\mathbb{Z}}, \qquad \mathbb{P} := \bigotimes_{k\in\mathbb{Z}} \tfrac{1}{2}(\delta_{-1} + \delta_{+1}),$$

and denote by $\omega = (\omega_k)_{k\in\mathbb{Z}} \in \Omega$ a typical element. Define the coordinate maps

$$\varepsilon_k : \Omega \to \{-1, +1\}, \qquad \varepsilon_k(\omega) := \omega_k.$$

Then $(\varepsilon_k)_{k\in\mathbb{Z}}$ is a sequence of independent Rademacher random variables, i.e.

$$\mathbb{P}(\varepsilon_k = \pm 1) = \tfrac{1}{2}, \quad \text{and the variables are independent.}$$

For each integer $n \ge 1$, define the random polynomial[1]

$$P_n(z) := \sum_{k=0}^{n-1} \varepsilon_k z^k, \qquad z \in \mathbb{C}.$$

[1]Our convention uses n coefficients (degree $n-1$), whereas Erdős's original formulation uses $n+1$ coefficients. Applying our results to P_{n+1} immediately yields the same Erdős formulation (See Corollary 9.6).

For $r > 0$, set

$$\sigma_n(r)^2 := \sum_{k=0}^{n-1} r^{2k},$$

and define the normalized process on the torus $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ by

$$Z_r(\theta) := \frac{P_n\big(re^{i\theta}\big)}{\sigma_n(r)}, \qquad \theta \in \mathbb{T}.$$

All integrals over $\mathbb{T}$ are taken with respect to the normalized Haar measure μ. We write

$$L_n(r) = \int_{\mathbb{T}} \log|Z_r(\theta)|\, d\mu(\theta), \qquad I_n = [1 - n^{-11/10}, 1 + n^{-11/10}].$$

We use the following notation throughout. For a random variable X,

$$\|X\|_{L^p(\Omega)} = (\mathbb{E}|X|^p)^{1/p}.$$

For a function g on the torus, $\|g\|_{L^p(\mathbb{T})} = (\int_{\mathbb{T}} |g|^p\, d\mu)^{1/p}$. The torus distance is

$$\mathrm{dist}_{\mathbb{T}}(u) := \min_{k\in\mathbb{Z}} |u - 2\pi k|,$$

where representatives are taken modulo 2π. In expressions such as $\min\{n, 1/\mathrm{dist}_{\mathbb{T}}(u)\}$, the value at $\mathrm{dist}_{\mathbb{T}}(u) = 0$ is understood to be n. For finite-dimensional real vectors, $|v|_2$ denotes the Euclidean norm, and $\|A\|_{\mathrm{op}}$ denotes the operator norm of a matrix. For $z \in \mathbb{C}$, $\mathrm{Re}\, z$ and $\mathrm{Im}\, z$ denote its real and imaginary parts.

Proposition 2.1 (Fixed-dimensional anti-concentration)**.** *Fix $d \in \mathbb{N}$ and $c_0, C_0 > 0$. Then there exists a constant $C_{\mathrm{KR}} = C_{\mathrm{KR}}(d, c_0, C_0) < \infty$ such that the following holds.*

Let

$$S = \sum_{k=1}^{N} \varepsilon_k v_k \in \mathbb{R}^d,$$

where $\varepsilon_1, \dots, \varepsilon_N$ are independent Rademacher signs and $v_1, \dots, v_N \in \mathbb{R}^d$ satisfy

$$\Gamma := \sum_{k=1}^{N} v_k v_k^\top \succeq c_0 N I_d, \qquad \max_{1\le k\le N} |v_k|_2 \le C_0.$$

Then for every axis-parallel box

$$Q = \prod_{j=1}^{d} [a_j, a_j + \ell_j] \subset \mathbb{R}^d \qquad (\ell_j \ge 0),$$

and every $x \in \mathbb{R}^d$,

$$\mathbb{P}(S \in x + Q) \le C_{\mathrm{KR}}\, N^{-d/2} \prod_{j=1}^{d} (\ell_j + 1).$$

In particular, if $\ell_j \le A t_j$ and $t_j \ge 1$ for all j, then

$$\mathbb{P}(S \in x + Q) \le C_{\mathrm{KR}}(A; d, c_0, C_0) \prod_{j=1}^{d} \frac{t_j}{\sqrt{N}}.$$

Proof. We use the Fourier convention

$$\widehat{g}(\xi) = \int_{\mathbb{R}^d} e^{-i\langle \xi, y\rangle} g(y)\, dy.$$

The characteristic function of S is

$$\varphi_S(\xi) = \mathbb{E} e^{i\langle \xi, S\rangle} = \prod_{k=1}^{N} \mathbb{E} e^{i\varepsilon_k \langle \xi, v_k\rangle} = \prod_{k=1}^{N} \cos\langle \xi, v_k\rangle.$$

Here we used the independence of the ε_k's.

Set

$$\eta := \frac{1}{2C_0}, \qquad \eta_1 := \frac{\eta}{\sqrt{d}}.$$

If $|\xi|_2 \le \eta$, then

$$|\langle \xi, v_k\rangle| \le |\xi|_2\, |v_k|_2 \le \eta C_0 = \frac{1}{2}.$$

For $|u| \le 1/2$, we have

$$|\cos u| = \cos u \le 1 - \frac{u^2}{2} + \frac{u^4}{24} \le 1 - \frac{11}{24} u^2 \le e^{-u^2/8}.$$

Hence, for $|\xi|_2 \le \eta$,

$$|\varphi_S(\xi)| \le \exp\left(-\frac{1}{8} \sum_{k=1}^{N} \langle \xi, v_k\rangle^2\right) = \exp\left(-\frac{1}{8}\, \xi^\top \Gamma \xi\right) \le \exp\Big(-\frac{c_0}{8} N\, |\xi|_2^2\Big).$$

Since $[-\eta_1, \eta_1]^d \subset B(0, \eta)$, we obtain

$$\int_{[-\eta_1,\eta_1]^d} |\varphi_S(\xi)|\, d\xi \le \int_{\mathbb{R}^d} e^{-(c_0/8)N|\xi|_2^2}\, d\xi.$$

By Fubini,

$$\int_{\mathbb{R}^d} e^{-(c_0/8)N|\xi|_2^2}\, d\xi = \prod_{j=1}^{d} \int_{\mathbb{R}} e^{-(c_0/8)Nu^2}\, du = \left(\frac{8\pi}{c_0 N}\right)^{d/2}.$$

Therefore

$$\int_{[-\eta_1,\eta_1]^d} |\varphi_S(\xi)|\, d\xi \le C_{d,c_0} N^{-d/2}. \tag{1}$$

Next we construct an L^1-majorant of $\mathbf{1}_Q$ whose Fourier transform is supported in $[-\eta_1, \eta_1]^d$.

Define

$$k(t) := \frac{\eta_1}{2\pi} \left(\frac{\sin(\eta_1 t/2)}{\eta_1 t/2}\right)^2.$$

This is the one-dimensional Fejér kernel at frequency η_1, so

$$k \ge 0, \qquad \widehat{k}(\tau) = \Big(1 - \frac{|\tau|}{\eta_1}\Big)_+, \qquad \int_{\mathbb{R}} k(t)\, dt = \widehat{k}(0) = 1.$$

Now set

$$K(y_1, \dots, y_d) := \prod_{j=1}^{d} k(y_j).$$

Then

$$K \ge 0, \qquad \int_{\mathbb{R}^d} K(y)\, dy = 1, \qquad \widehat{K}(\xi) = \prod_{j=1}^{d} \widehat{k}(\xi_j),$$

and therefore

$$\operatorname{supp} \widehat{K} \subset [-\eta_1, \eta_1]^d.$$

Let

$$m := \int_{[-1/2,1/2]^d} K(u)\, du.$$

Since K is continuous and $K(0) > 0$, we have $m > 0$. Note that m depends only on d and C_0.

Define

$$G_Q(y) := \frac{1}{m}\,(\mathbf{1}_{Q+[-1/2,1/2]^d} * K)(y) = \frac{1}{m}\int_{\mathbb{R}^d} \mathbf{1}_{Q+[-1/2,1/2]^d}(y-u)\, K(u)\, du.$$

If $y \in Q$ and $u \in [-1/2, 1/2]^d$, then $y - u \in Q + [-1/2, 1/2]^d$, hence

$$G_Q(y) \geq \frac{1}{m}\int_{[-1/2,1/2]^d} K(u)\, du = 1.$$

Thus

$$\mathbf{1}_Q \leq G_Q.$$

Also, by Fubini and $\int K = 1$,

$$\int_{\mathbb{R}^d} G_Q(y)\, dy = \frac{1}{m}\, \operatorname{vol}\big(Q + [-1/2, 1/2]^d\big).$$

If $Q = \prod_{j=1}^d [a_j, a_j + \ell_j]$, then

$$\operatorname{vol}\big(Q + [-1/2, 1/2]^d\big) = \prod_{j=1}^d (\ell_j + 1).$$

Hence

$$\int_{\mathbb{R}^d} |G_Q(y)|\, dy \leq C_{d,C_0} \prod_{j=1}^d (\ell_j + 1). \tag{2}$$

Moreover,

$$\widehat{G}_Q(\xi) = m^{-1}\, \widehat{\mathbf{1}_{Q+[-1/2,1/2]^d}}(\xi)\, \widehat{K}(\xi),$$

so $\operatorname{supp} \widehat{G}_Q \subset [-\eta_1, \eta_1]^d$. Since $G_Q \in L^1$,

$$|\widehat{G}_Q(\xi)| \leq \int_{\mathbb{R}^d} |G_Q(y)|\, dy \leq C_{d,C_0} \prod_{j=1}^d (\ell_j + 1).$$

Finally,

$$\mathbb{P}(S \in x + Q) \leq \mathbb{E}\, G_Q(S - x).$$

Because $\widehat{G}_Q$ is bounded and compactly supported, Fourier inversion gives

$$\mathbb{E}\, G_Q(S - x) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle \xi, x\rangle} \widehat{G}_Q(\xi) \varphi_S(\xi)\, d\xi.$$

Taking absolute values and using the support of $\widehat{G}_Q$,

$$\mathbb{P}(S \in x + Q) \leq (2\pi)^{-d} \sup_{\xi} |\widehat{G}_Q(\xi)| \int_{[-\eta_1,\eta_1]^d} |\varphi_S(\xi)|\, d\xi.$$

Combining this with (1) and (2), we get

$$\mathbb{P}(S \in x + Q) \le C_{d,c_0,C_0}\, N^{-d/2} \prod_{j=1}^{d} (\ell_j + 1).$$

If $\ell_j \le A t_j$ and $t_j \ge 1$, then

$$\ell_j + 1 \le (A+1) t_j,$$

so

$$\mathbb{P}(S \in x + Q) \le C_{\mathrm{KR}}(A; d, c_0, C_0) \prod_{j=1}^{d} \frac{t_j}{\sqrt{N}}.$$

This proves the proposition. □

We shall use the following theorem of Nazarov–Nishry–Sodin [2, Corollary 2.2].

Theorem 2.2 (Nazarov–Nishry–Sodin)**.** *There is an absolute constant $C_{\mathrm{NNS}} < \infty$ with the following property. Let*

$$f(\omega, \theta) = \sum_{k \in \mathbb{Z}} a_k \varepsilon_k(\omega) e^{ik\theta}, \qquad \sum_{k \in \mathbb{Z}} |a_k|^2 = 1,$$

be a normalized Rademacher Fourier series. Then, for every $p \ge 1$,

$$\int_{\Omega \times \mathbb{T}} |\log |f(\omega, \theta)||^p \, d\mathbb{P}(\omega)\, d\mu(\theta) \le (C_{\mathrm{NNS}} p)^{6p}.$$

To begin, we write $\log_+ x := \max\{\log x, 0\}$ for $x > 0$.

Lemma 2.3 (Logarithmic integrability for Z_r)**.** *For every fixed $0 < p < \infty$, there exists a constant $C_{\log}(p) < \infty$, depending only on p, such that*

$$\mathbb{E} \int_{\mathbb{T}} \log_+^p \frac{1}{|Z_r(\theta)|}\, d\mu(\theta) \le C_{\log}(p),$$

uniformly in n and $r > 0$. In particular, the same bound holds uniformly for $r \in I_n$.

Proof. Fix $n \ge 1$ and $r > 0$. Recall that

$$Z_r(\theta) = \frac{1}{\sigma_n(r)} \sum_{k=0}^{n-1} \varepsilon_k r^k e^{ik\theta},$$

where

$$\sigma_n(r)^2 = \sum_{k=0}^{n-1} r^{2k}.$$

It is convenient to rewrite Z_r in the form

$$Z_r(\theta) = \sum_{k=0}^{n-1} a_{k,r}\, \varepsilon_k e^{ik\theta}, \qquad a_{k,r} := \frac{r^k}{\sigma_n(r)}.$$

Since $r > 0$, the coefficients $a_{k,r}$ are nonnegative real numbers. By the definition of $\sigma_n(r)$,

$$\sum_{k=0}^{n-1} |a_{k,r}|^2 = \frac{1}{\sigma_n(r)^2} \sum_{k=0}^{n-1} r^{2k} = 1.$$

Thus, for each fixed $r > 0$, the random function $\theta \mapsto Z_r(\theta)$ is a normalized Rademacher Fourier series, in the sense that it has the form

$$f(\theta) = \sum_{k=0}^{n-1} a_k \varepsilon_k e^{ik\theta}$$

with

$$\sum_{k=0}^{n-1} |a_k|^2 = 1.$$

We now invoke Theorem 2.2. It gives, for every $q \geq 1$,

$$\mathbb{E} \int_{\mathbb{T}} |\log |f(\theta)||^q \, d\mu(\theta) \leq (C_{\mathrm{NNS}} q)^{6q}$$

for every normalized Rademacher Fourier series f. Applying this to $f = Z_r$, we obtain

$$\mathbb{E} \int_{\mathbb{T}} |\log |Z_r(\theta)||^q \, d\mu(\theta) \leq (C_{\mathrm{NNS}} q)^{6q}, \qquad q \geq 1. \tag{3}$$

Importantly, the right-hand side depends only on q, and not on n or r. This is the source of the desired uniformity.

Next we compare the positive logarithm of $1/|Z_r(\theta)|$ with $|\log |Z_r(\theta)||$. Recall that for any $x > 0$,

$$\log_+ \frac{1}{x} = \max\left\{ \log \frac{1}{x}, 0 \right\}.$$

If $x \leq 1$, then $\log_+(1/x) = \log(1/x) = -\log x$; if $x \geq 1$, then $\log_+(1/x) = 0$. In either case,

$$\log_+ \frac{1}{x} \leq |\log x|.$$

Applying this pointwise with $x = |Z_r(\theta)|$, we get

$$\log_+ \frac{1}{|Z_r(\theta)|} \leq |\log |Z_r(\theta)||.$$

Therefore, for every $p \geq 1$,

$$\log_+^p \frac{1}{|Z_r(\theta)|} \leq |\log |Z_r(\theta)||^p.$$

Integrating over $\mathbb{T}$ and taking expectation yields

$$\mathbb{E} \int_{\mathbb{T}} \log_+^p \frac{1}{|Z_r(\theta)|} \, d\mu(\theta) \leq \mathbb{E} \int_{\mathbb{T}} |\log |Z_r(\theta)||^p \, d\mu(\theta).$$

Now choose $q = p$ in (3). Since $p \geq 1$, this gives

$$\mathbb{E} \int_{\mathbb{T}} \log_+^p \frac{1}{|Z_r(\theta)|} \, d\mu(\theta) \leq (C_{\mathrm{NNS}} p)^{6p}.$$

Thus the claimed estimate holds for all $p \geq 1$, with

$$C_{\log}(p) := (C_{\mathrm{NNS}} p)^{6p}.$$

It remains to treat the case $0 < p < 1$. Set

$$X := \int_{\mathbb{T}} \log_+ \frac{1}{|Z_r(\theta)|} \, d\mu(\theta).$$

Since $X \geq 0$ and the function $t \mapsto t^p$ is concave on $[0, \infty)$ for $0 < p < 1$, Jensen's inequality gives

$$\int_{\mathbb{T}} \log_+^p \frac{1}{|Z_r(\theta)|}\, d\mu(\theta) \leq \left(\int_{\mathbb{T}} \log_+ \frac{1}{|Z_r(\theta)|}\, d\mu(\theta) \right)^p = X^p.$$

Taking expectations and using Jensen's inequality once more, now for the concave function $t \mapsto t^p$ on $[0, \infty)$, we obtain

$$\mathbb{E}X^p \leq (\mathbb{E}X)^p.$$

Hence

$$\mathbb{E} \int_{\mathbb{T}} \log_+^p \frac{1}{|Z_r(\theta)|}\, d\mu(\theta) \leq \left(\mathbb{E} \int_{\mathbb{T}} \log_+ \frac{1}{|Z_r(\theta)|}\, d\mu(\theta) \right)^p.$$

By the already proved $p = 1$ case,

$$\mathbb{E} \int_{\mathbb{T}} \log_+ \frac{1}{|Z_r(\theta)|}\, d\mu(\theta) \leq C_{\log}(1),$$

and therefore

$$\mathbb{E} \int_{\mathbb{T}} \log_+^p \frac{1}{|Z_r(\theta)|}\, d\mu(\theta) \leq C_{\log}(1)^p.$$

For $0 < p < 1$, we set $C_{\log}(p) := C_{\log}(1)^p$. This proves the desired bound in this range.

Combining the two cases, we conclude that for every fixed $p < \infty$,

$$\mathbb{E} \int_{\mathbb{T}} \log_+^p \frac{1}{|Z_r(\theta)|}\, d\mu(\theta) \leq C_{\log}(p),$$

with $C_{\log}(p)$ depending only on p, uniformly in n and $r > 0$. The final sentence follows immediately, since $I_n \subset (0, \infty)$. □

We shall also use the following consequence of Yakir's work [4, Lemma 1.3].

Theorem 2.4 (Yakir)**.** *There is an absolute constant $C_{\text{Yakir}} < \infty$ such that, for all sufficiently large n and uniformly for $r \in I_n$,*

$$\left| \mathbb{E}L_n(r) + \frac{\gamma}{2} \right| \leq C_{\text{Yakir}} \frac{(\log n)^2}{\sqrt{n}}.$$

Proof. Yakir's Lemma 1.3 is stated for the random Littlewood polynomial

$$P(z) = \sum_{k=0}^{n-1} \varepsilon_k z^k$$

and for the normalization

$$\sigma(r)^2 = \mathbb{E}|P(re^{i\theta})|^2 = \sum_{k=0}^{n-1} r^{2k}, \qquad \widetilde{P}(re^{i\theta}) = \frac{P(re^{i\theta})}{\sigma(r)}.$$

These are exactly the definitions of P_n, $\sigma_n(r)$, and $Z_r(\theta)$ used here. Taking $\ell = 1$ in Yakir's Lemma 1.3 gives, uniformly for

$$r \in [1 - n^{-11/10}, 1 + n^{-11/10}] = I_n,$$

$$\mathbb{E} \int_T \log |Z_r(\theta)|\, d\mu(\theta) = -\frac{\gamma}{2} + O\left(\frac{(\log n)^2}{\sqrt{n}} \right).$$

Since the integral on the left is $L_n(r)$, the claimed estimate follows. □

We shall also use Pisier's inequality on the discrete cube [3].

Theorem 2.5 (Pisier's inequality on the discrete cube)**.** *There is an absolute constant $C_{\mathrm{Pis}} < \infty$ such that, for every $n \geq 1$ and every real-valued function G on $\{-1,+1\}^n$,*

$$\|G - \mathbb{E}G\|_{L^4(\Omega)}^2 \leq C_{\mathrm{Pis}} \left\| \sum_{j=0}^{n-1} |G(\varepsilon) - G(\tau_j \varepsilon)|^2 \right\|_{L^2(\Omega)},$$

where $\tau_j\varepsilon$ is obtained from ε by flipping the j-th coordinate.

Remark 2.6. Proposition 2.1 is only used in fixed dimension. Theorem 2.2 is quoted in the normalized form needed below, Lemma 2.3 records the immediate specialization to the polynomials Z_r, and Theorem 2.5 is the only concentration input on the discrete cube.

Theorem 2.7 (Centered fourth moment)**.** *There exist absolute constants $C_{\mathrm{KL}}, \kappa_{\mathrm{KL}} < \infty$ such that, for all sufficiently large n and uniformly for $r \in I_n$,*

$$\boxed{\|L_n(r) - \mathbb{E}L_n(r)\|_{L^4(\Omega)} \leq C_{\mathrm{KL}} (\log n)^{\kappa_{\mathrm{KL}}} n^{-1/2}.}$$

Corollary 2.8 (Uncentered fourth moment)**.** *There exist absolute constants $C_{\mathrm{uKL}}, \kappa_{\mathrm{uKL}} < \infty$ such that, for all sufficiently large n and uniformly for $r \in I_n$,*

$$\boxed{\mathbb{E}\left|L_n(r) + \frac{\gamma}{2}\right|^4 \leq C_{\mathrm{uKL}} (\log n)^{\kappa_{\mathrm{uKL}}} n^{-2}.}$$

3. Elementary deterministic estimates

For $r \in I_n$, define

$$a_k = a_{k,r} := \frac{r^k}{\sigma_n(r)}, \qquad 0 \leq k \leq n-1,$$

where

$$\sigma_n(r)^2 = \sum_{k=0}^{n-1} r^{2k}.$$

We first record some elementary bounds on these coefficients.

Lemma 3.1 (Flatness of the coefficients $a_{k,r}$)**.** *There are absolute constants $C_\sigma, C_a < \infty$ such that, for all sufficiently large n and uniformly for $r \in I_n$,*

$$\left|\sigma_n(r)^2 - n\right| \leq C_\sigma n^{9/10}, \qquad \max_{0 \leq k \leq n-1} |a_k| \leq C_a n^{-1/2}, \qquad \sum_{k=0}^{n-1} a_k^2 = 1.$$

Proof. Write $r = 1 + x$, so $|x| \leq n^{-11/10}$. For all sufficiently large n, the elementary bound $|\log(1+x)| \leq 2|x|$ gives

$$|\log r| \leq 2n^{-11/10}.$$

Hence, for $0 \leq k \leq n-1$,

$$|2k \log r| \leq 4n^{-1/10} \leq 1.$$

Using $|e^y - 1| \leq e|y|$ for $|y| \leq 1$, we obtain

$$|r^{2k} - 1| = |e^{2k \log r} - 1| \leq 4e\, n^{-1/10}.$$

Summing this estimate over $0 \leq k < n$ yields

$$\left|\sigma_n(r)^2 - n\right| = \left|\sum_{k=0}^{n-1} (r^{2k} - 1)\right| \leq 4e\, n^{9/10}.$$

Thus the first claim holds, for example, with $C_\sigma = 4e$.

The same logarithmic estimate gives $r^k \le e^{2n^{-1/10}} \le e^2$ for all sufficiently large n. The first bound also gives $\sigma_n(r)^2 \ge n/2$ after increasing the threshold for n. Therefore

$$|a_k| = \frac{r^k}{\sigma_n(r)} \le \sqrt{2}e^2\, n^{-1/2},$$

uniformly in k and r. Thus the second claim holds with $C_a = \sqrt{2}e^2$, after enlarging the absolute constant if necessary.

Finally,

$$\sum_{k=0}^{n-1} a_k^2 = \frac{1}{\sigma_n(r)^2} \sum_{k=0}^{n-1} r^{2k} = 1$$

by the definition of $\sigma_n(r)$. □

We also need the following weighted Dirichlet estimate.

Lemma 3.2 (Weighted Dirichlet estimate)**.** *There is an absolute constant $C_{\mathrm{Dir}} < \infty$ such that, for all sufficiently large n, uniformly for $r \in I_n$ and $u \in \mathbb{T}$,*

$$\left|\sum_{k=0}^{n-1} r^{2k} e^{iku}\right| \le C_{\mathrm{Dir}} \min\left\{n, \frac{1}{\mathrm{dist}_{\mathbb{T}}(u)}\right\}.$$

Proof. Fix $r \in I_n$, and set

$$w_k := r^{2k}, \qquad D_m(u) := \sum_{j=0}^{m} e^{iju}.$$

Then

$$\sum_{k=0}^{n-1} r^{2k} e^{iku} = \sum_{k=0}^{n-1} w_k e^{iku}.$$

Since $r \in I_n$, there is a constant $A_1 > 0$ such that

$$|\log r| \le A_1 n^{-11/10}.$$

Hence for $0 \le k \le n-1$,

$$|2k \log r| \le 2nA_1 n^{-11/10} = 2A_1 n^{-1/10} \le 2A_1,$$

and therefore

$$|w_k| = e^{2k \log r} \le e^{2A_1} =: A_2.$$

In particular,

$$|w_{n-1}| \le A_2.$$

Next, since

$$w_{k+1} - w_k = r^{2k}(r^2 - 1),$$

we get

$$|w_{k+1} - w_k| \le A_2 |r^2 - 1|.$$

Summing over $0 \le k \le n-2$,

$$\sum_{k=0}^{n-2} |w_{k+1} - w_k| \le (n-1)A_2 |r^2 - 1|.$$

Because $r \in I_n$, there is $A_3 > 0$ such that

$$|r^2 - 1| \le A_3 n^{-11/10}.$$

Thus

$$\sum_{k=0}^{n-2} |w_{k+1} - w_k| \le A_2 A_3\, n^{-1/10}.$$

Now apply summation by parts:

$$\sum_{k=0}^{n-1} w_k e^{iku} = w_{n-1} D_{n-1}(u) + \sum_{m=0}^{n-2} (w_m - w_{m+1}) D_m(u).$$

Therefore

$$\left| \sum_{k=0}^{n-1} w_k e^{iku} \right| \le |w_{n-1}|\, |D_{n-1}(u)| + \sum_{m=0}^{n-2} |w_m - w_{m+1}|\, |D_m(u)|.$$

For every $m \ge 0$, the Dirichlet kernel satisfies

$$|D_m(u)| \le A_4 \min\left\{ m+1, \frac{1}{\mathrm{dist}_{\mathbb{T}}(u)} \right\},$$

with an absolute constant $A_4 > 0$. Since $m + 1 \le n$ for $m \le n - 1$, we obtain

$$|D_m(u)| \le A_4 \min\left\{ n, \frac{1}{\mathrm{dist}_{\mathbb{T}}(u)} \right\}.$$

Hence

$$\left| \sum_{k=0}^{n-1} w_k e^{iku} \right| \le A_4 \min\left\{ n, \frac{1}{\mathrm{dist}_{\mathbb{T}}(u)} \right\} \left(|w_{n-1}| + \sum_{m=0}^{n-2} |w_m - w_{m+1}| \right).$$

Using the bounds above,

$$|w_{n-1}| + \sum_{m=0}^{n-2} |w_m - w_{m+1}| \le A_2 + A_2 A_3 n^{-1/10} \le A_2(1 + A_3).$$

Therefore

$$\left| \sum_{k=0}^{n-1} r^{2k} e^{iku} \right| \le A_4 A_2 (1 + A_3) \min\left\{ n, \frac{1}{\mathrm{dist}_{\mathbb{T}}(u)} \right\}.$$

This proves the claim with

$$C_{\mathrm{Dir}} := A_4 A_2 (1 + A_3). \qquad \square$$

For $\theta, \phi \in \mathbb{T}$, we set

$$d_0(\theta) := \min\{\mathrm{dist}_{\mathbb{T}}(\theta)\,,\ \mathrm{dist}_{\mathbb{T}}(\theta - \pi)\}, \qquad d_{\pm}(\theta, \phi) := \min\{\mathrm{dist}_{\mathbb{T}}(\theta - \phi)\,,\ \mathrm{dist}_{\mathbb{T}}(\theta + \phi)\},$$

and we define

$$f_r(\theta) := P_n(re^{i\theta}) = \sum_{k=0}^{n-1} \varepsilon_k r^k e^{ik\theta}.$$

Lemma 3.3 (Mixed covariance lower bound)**.** *Fix $m \le 6$. There exist constants $c_m, C_m > 0$ and $A_m \ge 1$, depending only on m, such that the following holds. Let n be sufficiently large, let $r \in I_n$, and let $0 < \rho \le 1$ satisfy*

$$\rho \ge \frac{A_m}{n}.$$

Let

$$\vartheta_1, \dots, \vartheta_q, \phi_1, \dots, \phi_a \in \mathbb{T}$$

with $2q+a \le 2m$, and assume:

$$\begin{aligned} d_0(\vartheta_i) &\ge \rho && (1 \le i \le q),\\ d_\pm(\vartheta_i,\vartheta_j) &\ge \rho && (1 \le i \ne j \le q),\\ d_\pm(\vartheta_i,\phi_\ell) &\ge \rho && (1 \le i \le q,\ 1 \le \ell \le a). \end{aligned}$$

Assume moreover that for each ℓ there exists $\psi_\ell \in \{0,\pi\}$ such that

$$\mathrm{dist}_{\mathbb{T}}(\phi_\ell - \psi_\ell) \le \rho,$$

and that the poles $\psi_1,\dots,\psi_a$ are pairwise distinct. Define

$$Y = \big(\mathrm{Re}\, f_r(\vartheta_1), \mathrm{Im}\, f_r(\vartheta_1), \dots, \mathrm{Re}\, f_r(\vartheta_q), \mathrm{Im}\, f_r(\vartheta_q), \mathrm{Re}\, f_r(\phi_1), \dots, \mathrm{Re}\, f_r(\phi_a)\big) \in \mathbb{R}^{2q+a}.$$

Then

$$c_m n I_{2q+a} \preceq \mathrm{Cov}(Y) \preceq C_m n I_{2q+a}.$$

Moreover, if $Y = \sum_{k=0}^{n-1} \varepsilon_k v_k$, then

$$\max_{0\le k<n} |v_k|_2 \le C_m.$$

Proof. Write

$$Y = \sum_{k=0}^{n-1} \varepsilon_k v_k,$$

where

$$v_k = \begin{pmatrix} r^k\cos(k\vartheta_1),\ r^k\sin(k\vartheta_1), \dots, r^k\cos(k\vartheta_q),\ r^k\sin(k\vartheta_q),\\ r^k\cos(k\phi_1), \dots, r^k\cos(k\phi_a)\end{pmatrix} \in \mathbb{R}^{2q+a}.$$

Since the variables ε_k are independent, centered, and satisfy $\mathbb{E}[\varepsilon_k^2] = 1$, we have

$$\mathrm{Cov}(Y) = \sum_{k=0}^{n-1} v_k v_k^\top.$$

We divide the proof into four steps.

Step 1: preliminary bounds on the weights. As in Lemma 3.1, $|\log r| \le 2n^{-11/10}$ for all sufficiently large n. Hence $|2k\log r| \le 4n^{-1/10} \le 1$ for $0 \le k < n$. We therefore fix absolute constants

$$c_w := e^{-1}, \qquad C_w := e,$$

so that

$$c_w \le r^{2k} \le C_w \qquad (0 \le k < n), \qquad \sum_{k=0}^{n-1} r^{2k} \ge c_w n.$$

Step 2: lower bounds for the diagonal blocks.

Step 2a: non-pole points. Fix $i \in \{1,\dots,q\}$, and set

$$X_i := \big(\mathrm{Re}\, f_r(\vartheta_i),\ \mathrm{Im}\, f_r(\vartheta_i)\big) \in \mathbb{R}^2.$$

Then

$$X_i = \sum_{k=0}^{n-1} \varepsilon_k \begin{pmatrix} r^k\cos(k\vartheta_i)\\ r^k\sin(k\vartheta_i)\end{pmatrix},$$

so

$$\mathrm{Cov}(X_i) = \sum_{k=0}^{n-1} r^{2k} \begin{pmatrix} \cos^2(k\vartheta_i) & \cos(k\vartheta_i)\sin(k\vartheta_i)\\ \cos(k\vartheta_i)\sin(k\vartheta_i) & \sin^2(k\vartheta_i)\end{pmatrix}.$$

Using

$$\cos^2 x = \frac{1+\cos 2x}{2}, \qquad \sin^2 x = \frac{1-\cos 2x}{2}, \qquad \sin x \cos x = \frac{\sin 2x}{2},$$

we may write

$$\mathrm{Cov}(X_i) = \frac{1}{2}\Big(\sum_{k=0}^{n-1} r^{2k}\Big) I_2 + E_i,$$

where each entry of E_i is a linear combination of

$$\sum_{k=0}^{n-1} r^{2k}\cos(2k\vartheta_i), \qquad \sum_{k=0}^{n-1} r^{2k}\sin(2k\vartheta_i).$$

Hence, by Lemma 3.2, there is an absolute constant $C_{\mathrm{diag}} := 2C_{\mathrm{Dir}}$ such that

$$\|E_i\|_{\mathrm{op}} \le C_{\mathrm{diag}} \min\left\{n, \frac{1}{\mathrm{dist}_{\mathbb{T}}(2\vartheta_i)}\right\}.$$

Since $\rho \le 1 < \pi/2$, and

$$d_0(\vartheta_i) = \min\{\mathrm{dist}_{\mathbb{T}}(\vartheta_i)\,, \mathrm{dist}_{\mathbb{T}}(\vartheta_i - \pi)\},$$

we have

$$\mathrm{dist}_{\mathbb{T}}(2\vartheta_i) = 2d_0(\vartheta_i) \ge 2\rho.$$

Therefore

$$\|E_i\|_{\mathrm{op}} \le \frac{C_{\mathrm{diag}}}{2\rho}.$$

It follows that

$$\mathrm{Cov}(X_i) \;\succeq\; \left(\frac{1}{2}\sum_{k=0}^{n-1} r^{2k} - \frac{C_{\mathrm{diag}}}{2\rho}\right) I_2 \;\succeq\; \left(\frac{c_w}{2}n - \frac{C_{\mathrm{diag}}}{2\rho}\right) I_2.$$

If $A_m \ge 2C_{\mathrm{diag}}/c_w$, then $\rho \ge A_m/n$ implies

$$\mathrm{Cov}(X_i) \succeq c_1 n I_2, \qquad c_1 := c_w/4.$$

We also record that the off-diagonal entry inside this same non-pole 2×2 block is small. Indeed,

$$\big|\mathrm{Cov}\big(\mathrm{Re}\, f_r(\vartheta_i), \mathrm{Im}\, f_r(\vartheta_i)\big)\big| \le \|E_i\|_{\mathrm{op}} \le \frac{C_{\mathrm{diag}}}{2\rho} \le \frac{C_{\mathrm{diag}}}{2A_m} n.$$

Since $C_{\mathrm{diag}} = 2C_{\mathrm{Dir}}$, this is at most

$$\frac{C_{\mathrm{Dir}}}{A_m} n.$$

Step 2b: near-pole points. Fix $\ell \in \{1, \dots, a\}$, and set

$$W_\ell := \mathrm{Re}\, f_r(\phi_\ell) = \sum_{k=0}^{n-1} \varepsilon_k r^k \cos(k\phi_\ell).$$

Then

$$\mathrm{Var}(W_\ell) = \sum_{k=0}^{n-1} r^{2k}\cos^2(k\phi_\ell).$$

We prove that $\mathrm{Var}(W_\ell) \ge c_2 n$, with $c_2 > 0$ depending only on m.

Choose $\psi_\ell \in \{0, \pi\}$ and a representative $\eta_\ell \in [-1, 1]$ such that

$$\phi_\ell = \psi_\ell + \eta_\ell \pmod{2\pi}, \qquad |\eta_\ell| = \mathrm{dist}_{\mathbb{T}}(\phi_\ell - \psi_\ell) \le \rho.$$

Since $|\cos(k\psi_\ell)| = 1$, we have

$$|\cos(k\phi_\ell)| = |\cos(k\eta_\ell)|.$$

Let $K_{\text{pole}} \geq 1$ be a sufficiently large absolute constant.

If $|\eta_\ell| \leq K_{\text{pole}}/n$, then for all $0 \leq k \leq \lfloor n/(3K_{\text{pole}})\rfloor$,

$$|k\eta_\ell| \leq \frac{1}{3}, \qquad \cos^2(k\eta_\ell) \geq \frac{1}{4}.$$

Using the lower bound $r^{2k} \geq c_w$ from Step 1, we get

$$\operatorname{Var}(W_\ell) \geq \frac{1}{4} \sum_{0\leq k\leq n/(3K_{\text{pole}})} r^{2k} \geq c(K_{\text{pole}})n.$$

It remains to consider $|\eta_\ell| > K_{\text{pole}}/n$. In this case, since $|\eta_\ell| \leq 1$, $\operatorname{dist}_{\mathbb{T}}(2\phi_\ell) = \operatorname{dist}_{\mathbb{T}}(2\eta_\ell) = 2|\eta_\ell|$. By Lemma 3.2 and Step 1,

$$\operatorname{Var}(W_\ell) \geq \frac{c_w}{2}n - \frac{C_{\text{Dir}}}{2\ \operatorname{dist}_T(2\phi_\ell)}.$$

Since in the present case

$$\operatorname{dist}_T(2\phi_\ell) = 2|\eta_\ell| > \frac{2K_{\text{pole}}}{n},$$

we obtain

$$\operatorname{Var}(W_\ell) \geq \frac{c_w}{2}n - \frac{C_{\text{Dir}}}{4K_{\text{pole}}}n.$$

Choosing K_{pole} sufficiently large gives

$$\operatorname{Var}(W_\ell) \geq \frac{c_w}{4}n.$$

Combining this with the previous case, we obtain

$$\operatorname{Var}(W_\ell) \geq c_2 n$$

with

$$c_2 := \min\{c(K_{\text{pole}}),\, c_w/4\} > 0.$$

Step 3: bounds for the off-diagonal entries. We now bound the cross-block off-diagonal entries of $\operatorname{Cov}(Y)$. The only off-diagonal entries not covered in this step are the entries inside the same non-pole 2×2 block, and these were already controlled at the end of Step 2a. Every entry of $\operatorname{Cov}(Y)$ is a finite linear combination of sums of the form

$$\sum_{k=0}^{n-1} r^{2k} e^{ik\omega},$$

where ω is one of

$$\pm(\vartheta_i - \vartheta_j), \qquad \pm(\vartheta_i + \vartheta_j), \qquad \pm(\vartheta_i - \phi_\ell), \qquad \pm(\vartheta_i + \phi_\ell), \qquad \pm(\phi_\ell - \phi_{\ell'}), \qquad \pm(\phi_\ell + \phi_{\ell'}).$$

We check that in each off-diagonal case,

$$\operatorname{dist}_{\mathbb{T}}(\omega) \geq \rho.$$

For two non-pole points ϑ_i, ϑ_j, this is exactly the assumption

$$d_\pm(\vartheta_i, \vartheta_j) \geq \rho.$$

For one non-pole point and one near-pole point, this is exactly the assumption

$$d_\pm(\vartheta_i, \phi_\ell) \geq \rho.$$

For two near-pole points $\phi_\ell, \phi_{\ell'}$ with $\ell \neq \ell'$, the associated poles $\psi_\ell, \psi_{\ell'}$ are distinct, hence one equals 0 and the other equals π. Writing

$$\phi_\ell = \psi_\ell + \eta_\ell, \qquad \phi_{\ell'} = \psi_{\ell'} + \eta_{\ell'}$$

with $|\eta_\ell|, |\eta_{\ell'}| \leq \rho$, we have

$$\mathrm{dist}_{\mathbb{T}}((\phi_\ell - \phi_{\ell'}) - \pi) \leq 2\rho, \qquad \mathrm{dist}_{\mathbb{T}}((\phi_\ell + \phi_{\ell'}) - \pi) \leq 2\rho,$$

and therefore

$$\mathrm{dist}_{\mathbb{T}}(\phi_\ell - \phi_{\ell'}) \geq \pi - 2\rho, \qquad \mathrm{dist}_{\mathbb{T}}(\phi_\ell + \phi_{\ell'}) \geq \pi - 2\rho.$$

Since $0 < \rho \leq 1 < \pi/3$,

$$\mathrm{dist}_{\mathbb{T}}(\phi_\ell - \phi_{\ell'}) \geq \rho, \qquad \mathrm{dist}_{\mathbb{T}}(\phi_\ell + \phi_{\ell'}) \geq \rho.$$

Thus in every off-diagonal case, Lemma 3.2 gives

$$\left| \sum_{k=0}^{n-1} r^{2k} e^{ik\omega} \right| \leq C_{\mathrm{Dir}} \min\left\{ n, \frac{1}{\mathrm{dist}_{\mathbb{T}}(\omega)} \right\} \leq \frac{C_{\mathrm{Dir}}}{\rho} \leq \frac{C_{\mathrm{Dir}}}{A_m} n.$$

Hence each off-diagonal entry of $\mathrm{Cov}(Y)$ has absolute value at most

$$\frac{C_{\mathrm{Dir}}}{A_m} n.$$

Step 4: conclusion by Gershgorin. Each diagonal entry corresponding to a non-pole coordinate is bounded below by $c_1 n$, and each diagonal entry corresponding to a near-pole coordinate is bounded below by $c_2 n$. Set

$$c_* := \min\{c_1, c_2\}.$$

By Step 2a, the off-diagonal entry inside each non-pole 2×2 block has absolute value at most $C_{\mathrm{Dir}} n / A_m$. By Step 3, every cross-block off-diagonal entry has absolute value at most $C_{\mathrm{Dir}} n / A_m$. Therefore every off-diagonal entry of $\mathrm{Cov}(Y)$ has absolute value at most $C_{\mathrm{Dir}} n / A_m$.

Since $2q + a \leq 2m$, each row has at most $2m - 1$ off-diagonal entries. Therefore the sum of the absolute values of the off-diagonal entries in any row is at most

$$(2m-1) \frac{C_{\mathrm{Dir}}}{A_m} n.$$

Choose A_m sufficiently large so that

$$(2m-1) \frac{C_{\mathrm{Dir}}}{A_m} \leq \frac{c_*}{2}.$$

Then every Gershgorin disc is contained in $[c_* n/2, \infty)$, and hence

$$\mathrm{Cov}(Y) \succeq \frac{c_*}{2} n \, I_{2q+a}.$$

This gives the desired lower bound with $c_m := c_*/2$.

For the upper bound, note that each coordinate of v_k has absolute value at most r^k, so

$$|v_k|_2^2 \leq (2q+a) r^{2k} \leq 2m C_w.$$

After enlarging C_m, this pointwise bound is part of the same constant in the statement. Hence

$$\mathrm{Cov}(Y) = \sum_{k=0}^{n-1} v_k v_k^\top \preceq \sum_{k=0}^{n-1} |v_k|_2^2 \, I_{2q+a} \preceq C_m n I_{2q+a}.$$

The same pointwise estimate gives

$$\max_{0\le k<n} |v_k|_2 \le C_m.$$

This completes the proof. □

4. Occupation estimates

For fixed n and r, write

$$M_{n,r}(u) = \mu\{\theta \in \mathbb{T} : |f_r(\theta)| \le u\}.$$

To reduce notation inside this section, we write $M(u) = M_{n,r}(u)$.

Lemma 4.1 (Cluster bookkeeping)**.** *Fix $p \ge 1$. There exists $\rho_p > 0$ such that the following holds for all $0 < \rho \le \rho_p$. Given $\theta = (\theta_1, \dots, \theta_p) \in \mathbb{T}^p$, define a graph with vertex set*

$$V := \{1, \dots, p\} \cup \{0_*, \pi_*\},$$

where:

- *for $i \ne j$, connect i and j if $d_\pm(\theta_i, \theta_j) \le \rho$;*
- *connect i to 0_* if $\mathrm{dist}_{\mathbb{T}}(\theta_i) \le \rho$;*
- *connect i to π_* if $\mathrm{dist}_{\mathbb{T}}(\theta_i - \pi) \le \rho$.*

Remove isolated pole vertices. Let the resulting graph have q connected components containing ordinary vertices and no pole, and a connected components containing ordinary vertices and exactly one pole. Then:

(1) $a \le 2$;
(2) one may choose representatives $\vartheta_1, \dots, \vartheta_q, \phi_1, \dots, \phi_a$ (one from each component) such that

$$\begin{aligned} d_0(\vartheta_i) &> \rho, \\ d_\pm(\vartheta_i, \vartheta_j) &> \rho \quad (i \ne j), \\ d_\pm(\vartheta_i, \phi_\ell) &> \rho, \end{aligned}$$

and for each ℓ, there exists $\psi_\ell \in \{0, \pi\}$ with

$$\mathrm{dist}_{\mathbb{T}}(\phi_\ell - \psi_\ell) \le \rho,$$

where the ψ_ℓ are pairwise distinct.

For each fixed graph pattern with q components not containing a pole, the set of $\theta \in \mathbb{T}^p$ realizing that pattern has measure at most

$$C_{\mathrm{clust}}(p)\, \rho^{p-q},$$

where $C_{\mathrm{clust}}(p) < \infty$ depends only on p.

Proof. We divide the proof into three steps. We shall use that $d_\pm$ is the quotient metric on $\mathbb{T}/\{\theta \sim -\theta\}$. In particular it satisfies the triangle inequality, and $d_\pm(0, \pi) = \pi$.

Step 1: structure of components.

We first show that each connected component contains at most one pole, and hence $a \le 2$.

Suppose a component contains both 0_* and π_*. Then there exists a path from 0_* to π_* of length at most $p + 1$, where each edge has $d_\pm$-length at most ρ. Hence, in the metric $d_\pm$, the distance between 0 and π is at most $(p + 1)\rho$. Since this distance equals π, this is impossible if $(p + 1)\rho < \pi/2$. Choosing ρ_p sufficiently small ensures this does not occur.

Thus each component contains at most one pole. Moreover, two distinct components cannot contain the same pole, since they would then be connected through that pole vertex. Therefore $a \le 2$.

Step 2: choice of representatives.

For each component not containing a pole, choose any vertex as its representative ϑ_i.

For each component containing a pole, choose a representative ϕ_ℓ among the ordinary vertices adjacent to that pole. Such a vertex exists because the component contains ordinary vertices, and any path from the pole to an ordinary vertex must begin with an edge from the pole.

We verify the required properties:

(i) If $d_0(\vartheta_i) \le \rho$, then ϑ_i would be connected to a pole, contradicting the assumption that its component contains no pole. Hence $d_0(\vartheta_i) > \rho$.

(ii) If $d_\pm(\vartheta_i, \vartheta_j) \le \rho$ for $i \ne j$, then the corresponding components would be connected by an edge, contradicting their distinctness.

(iii) If $d_\pm(\vartheta_i, \phi_\ell) \le \rho$, then the components would again be connected, a contradiction.

(iv) For each ϕ_ℓ, by construction there exists $\psi_\ell \in \{0, \pi\}$ such that $\mathrm{dist}_{\mathbb{T}}(\phi_\ell - \psi_\ell) \le \rho$. Since different components cannot contain the same pole, the poles ψ_ℓ are pairwise distinct.

Step 3: volume bound.

Fix a graph pattern and ignore all non-edge constraints, as this only enlarges the set. For one torus variable, each condition of the form $d_\pm(\theta, \eta) \le \rho$ confines θ to the union of at most two intervals of μ-measure at most ρ/π each, and each condition $\mathrm{dist}_{\mathbb{T}}(\theta - \psi) \le \rho$ confines θ to one interval of μ-measure at most ρ/π. Thus every tree edge contributes at most $2\rho/\pi$ to the product measure.

Consider first a component not containing a pole, with s ordinary vertices. Choose a spanning tree rooted at the representative. The root is unconstrained, while the remaining $s-1$ vertices are constrained by tree edges, so this component contributes at most $(2\rho/\pi)^{s-1}$.

Next consider a component containing a pole, with s ordinary vertices. Choose a spanning tree of the full component, rooted at the pole. Every ordinary vertex is constrained along exactly one tree edge, so this component contributes at most $(2\rho/\pi)^s$.

Multiplying over all components gives a factor $(2\rho/\pi)^{p-q}$. The number of graph patterns on $p+2$ labelled vertices is finite; absorbing this number and the harmless factors above into $C_{\mathrm{clust}}(p)$ proves the volume bound. □

Proposition 4.2 (Occupation moments)**.** *For each integer $1 \le p \le 6$ and each $K_u \ge 1$, there is a constant $C_{\mathrm{occ}}(p, K_u) < \infty$ such that, for all sufficiently large n, uniformly for $r \in I_n$ and $0 \le u \le K_u\sqrt{n}$,*

$$\|M_{n,r}(u)\|_{L^p(\Omega)} \le C_{\mathrm{occ}}(p, K_u)\frac{(1+u)^2}{n}.$$

Proof. Set

$$t := 1 + u, \qquad \rho := \frac{A_p t}{n},$$

where $A_p = A_p(p, K_u)$ is a large constant to be chosen. Since $u \le K_u\sqrt{n}$, we have

$$\rho \le C_\rho(p, K_u) n^{-1/2}, \qquad C_\rho(p, K_u) := A_p(1 + K_u),$$

so in particular $\rho \le \rho_p$ for all sufficiently large n. Thus Lemma 4.1 applies.

We start from

$$\mathbb{E}M(u)^p = \int_{\mathbb{T}^p} \mathbb{P}\big(|f_r(\theta_j)| \le u,\ 1 \le j \le p\big) \prod_{j=1}^{p} d\mu(\theta_j). \tag{4}$$

We decompose $\mathbb{T}^p$ according to the graph patterns of Lemma 4.1. Let $N_{\mathrm{pat}}(p)$ denote the finite number of graph patterns on the labelled vertex set $\{1,\dots,p,0_*,\pi_*\}$. Fix one such pattern, and let q be the number of components not containing a pole and a the number of components containing a pole. By Lemma 4.1, the set of $\theta \in \mathbb{T}^p$ realizing this pattern has measure at most

$$C_{\mathrm{clust}}(p)\rho^{p-q}. \tag{5}$$

Choose representatives

$$\vartheta_1,\dots,\vartheta_q,\phi_1,\dots,\phi_a$$

as in Lemma 4.1. If $|f_r(\theta_j)| \le u$ for all j, then in particular

$$|f_r(\vartheta_i)| \le u \qquad (1 \le i \le q),$$

and

$$|\operatorname{Re} f_r(\phi_\ell)| \le u \qquad (1 \le \ell \le a),$$

since $|f_r(\phi_\ell)| \le u$ implies $|\operatorname{Re} f_r(\phi_\ell)| \le u$.

Now take A_p sufficiently large. Since every component contains at least one ordinary vertex, $q + a \le p$. Therefore

$$2q + a \le 2(q + a) \le 2p.$$

Thus Lemma 3.3 applies with $m = p$, since $p \le 6$. Moreover, $\rho = A_p t/n$ and the representatives satisfy the hypotheses of Lemma 3.3. Therefore the vector

$$Y = \big(\operatorname{Re} f_r(\vartheta_1), \operatorname{Im} f_r(\vartheta_1), \dots, \operatorname{Re} f_r(\vartheta_q), \operatorname{Im} f_r(\vartheta_q), \operatorname{Re} f_r(\phi_1), \dots, \operatorname{Re} f_r(\phi_a)\big) \in \mathbb{R}^{2q+a}$$

has covariance bounded below by $c_{\mathrm{mix}}(p)nI$, and in the representation

$$Y = \sum_{k=0}^{n-1} \varepsilon_k v_k$$

the coefficient vectors satisfy $\max_k |v_k|_2 \le C_{\mathrm{mix}}(p)$.

The representative event is contained in an axis-parallel box in $\mathbb{R}^{2q+a}$ with side lengths at most $2t$ in each coordinate. Hence Proposition 2.1 yields

$$\mathbb{P}(\text{representative events}) \le C'_{\mathrm{occ}}(p, K_u)\left(\frac{t}{\sqrt{n}}\right)^{2q+a} = C'_{\mathrm{occ}}(p, K_u)\left(\frac{t^2}{n}\right)^q\left(\frac{t}{\sqrt{n}}\right)^a. \tag{6}$$

Combining (5) and (6), the contribution of this pattern to (4) is at most

$$C'_{\mathrm{occ}}(p, K_u)\left(\frac{t}{n}\right)^{p-q}\left(\frac{t^2}{n}\right)^q\left(\frac{t}{\sqrt{n}}\right)^a = C'_{\mathrm{occ}}(p, K_u)\frac{t^{p+q+a}}{n^{p+a/2}}.$$

Since each of the $q + a$ components contains at least one ordinary vertex, we have

$$q + a \le p.$$

Therefore, using $t \ge 1$,

$$\frac{t^{p+q+a}}{n^{p+a/2}} \le \frac{t^{2p}}{n^p}.$$

Summing over the $N_{\mathrm{pat}}(p)$ graph patterns and absorbing this finite factor into the same named constant, we obtain

$$\mathbb{E}M(u)^p \le C'_{\mathrm{occ}}(p, K_u)\frac{(1+u)^{2p}}{n^p}.$$

Taking p-th roots gives

$$\|M(u)\|_{L^p(\Omega)} \le C_{\mathrm{occ}}(p, K_u)\frac{(1+u)^2}{n}.$$

□

5. Resolvent bounds

Fix $B \geq 1$. In this section, statement-level constants are written with their B-dependence explicitly. Set

$$\Lambda = (\log n)^B, \qquad \alpha = \frac{\Lambda}{\sqrt{n}},$$

and define

$$R_\alpha = \int_{\mathbb{T}} \frac{d\mu(\theta)}{|Z_r(\theta)|^2 + \alpha^2}.$$

For the proof below, write $M(u) = M_{n,r}(u)$.

Proposition 5.1 (Resolvent moments)**.** *For $q = 2, 4$, there are constants $C_{\mathrm{res}}(q, B), \kappa_{\mathrm{res}}(q, B) < \infty$ such that, for all sufficiently large n and uniformly for $r \in I_n$,*

$$\boxed{\|R_\alpha\|_{L^q(\Omega)} \leq C_{\mathrm{res}}(q, B)(\log n)^{\kappa_{\mathrm{res}}(q,B)}.}$$

Proof. By Lemma 3.1, after increasing the threshold for n if necessary,

$$\frac{1}{2}n \leq \sigma_n(r)^2 \leq 2n, \qquad \frac{1}{2}\Lambda \leq A := \sigma_n(r)\alpha \leq 2\Lambda.$$

The layer-cake identity gives

$$\frac{1}{|f_r(\theta)|^2 + A^2} = \int_{|f_r(\theta)|}^{\infty} \frac{2u}{(u^2 + A^2)^2}\, du.$$

After integrating over $\mathbb{T}$,

$$\int_{\mathbb{T}} \frac{d\mu(\theta)}{|f_r(\theta)|^2 + A^2} = \int_0^\infty \frac{2u}{(u^2 + A^2)^2} M(u)\, du.$$

Since $R_\alpha = \sigma_n(r)^2 \int_{\mathbb{T}} (|f_r|^2 + A^2)^{-1}\, d\mu$, we obtain

$$R_\alpha \leq 2n \int_0^\infty \frac{2u}{(u^2 + A^2)^2} M(u)\, du.$$

Minkowski's integral inequality therefore yields

$$\|R_\alpha\|_{L^q(\Omega)} \leq 2n \int_0^\infty \frac{2u}{(u^2 + A^2)^2} \|M(u)\|_{L^q(\Omega)}\, du.$$

We split the integral at $u = \sqrt{n}$. Since $0 \leq M(u) \leq 1$, the tail contribution is bounded by

$$2n \int_{\sqrt{n}}^\infty \frac{2u}{(u^2 + A^2)^2}\, du = \frac{2n}{n + A^2} \leq 2.$$

On $0 \leq u \leq \sqrt{n}$, Proposition 4.2 with $K_u = 1$ gives

$$\|M(u)\|_{L^q(\Omega)} \leq C_{\mathrm{occ}}(q, 1) \frac{(1 + u)^2}{n}.$$

Thus

$$\|R_\alpha\|_{L^q(\Omega)} \leq 2C_{\mathrm{occ}}(q, 1) I_A + 2, \qquad I_A := \int_0^{\sqrt{n}} \frac{2u(1 + u)^2}{(u^2 + A^2)^2}\, du.$$

For all sufficiently large n, $A \geq 1$. On $0 \leq u \leq A$,

$$\int_0^A \frac{2u(1 + u)^2}{(u^2 + A^2)^2}\, du \leq \frac{1}{A^4} \int_0^A 2u(1 + u)^2\, du \leq 3.$$

On $A \le u \le \sqrt{n}$, the interval is empty if $A > \sqrt{n}$; otherwise $u \ge 1$, so $(1+u)^2 \le 4u^2$ and

$$\int_A^{\sqrt{n}} \frac{2u(1+u)^2}{(u^2+A^2)^2}\,du \le \int_A^{\sqrt{n}} \frac{8}{u}\,du \le 4\log n.$$

Consequently $I_A \le 7\log n$ for all sufficiently large n. We conclude that

$$\|R_\alpha\|_{L^q(\Omega)} \le \big(14C_{\mathrm{occ}}(q,1)+2\big)\log n.$$

This proves the proposition, for example with

$$C_{\mathrm{res}}(q,B) := 14C_{\mathrm{occ}}(q,1)+2, \qquad \kappa_{\mathrm{res}}(q,B) := 1.$$

□

6. Finite-difference carré-du-champ

For $x = (x_0,\dots,x_{n-1}) \in [-1,1]^n$, put

$$Z_x(\theta) = \sum_{k=0}^{n-1} a_k x_k e^{ik\theta}, \qquad F_\alpha(x) = \frac{1}{2}\int_{\mathbb{T}} \log\big(|Z_x(\theta)|^2+\alpha^2\big)\,d\mu(\theta).$$

For a sign vector ε, write $F_\alpha = F_\alpha(\varepsilon)$. Let $\tau_j\varepsilon$ be obtained from ε by flipping the j-th sign and define

$$\Delta_j F_\alpha = F_\alpha(\varepsilon) - F_\alpha(\tau_j\varepsilon).$$

Lemma 6.1 (Continuous gradient)**.** *There is an absolute constant $C_{\mathrm{grad}} < \infty$ such that, for every $x \in [-1,1]^n$,*

$$\sum_{j=0}^{n-1} |\partial_{x_j} F_\alpha(x)|^2 \le \frac{C_{\mathrm{grad}}}{n}\int_{\mathbb{T}} \frac{d\mu(\theta)}{|Z_x(\theta)|^2+\alpha^2}.$$

Moreover, for each j,

$$|\partial^2_{x_jx_j} F_\alpha(x)| \le C_{\mathrm{grad}} a_j^2 \int_{\mathbb{T}} \frac{d\mu(\theta)}{|Z_x(\theta)|^2+\alpha^2}.$$

Proof. Let

$$\Phi(z) = \tfrac{1}{2}\log(|z|^2+\alpha^2), \qquad H_x(\theta) := \frac{\overline{Z_x(\theta)}}{|Z_x(\theta)|^2+\alpha^2}.$$

Since $\partial_{x_j} Z_x(\theta) = a_j e^{ij\theta}$, the chain rule gives

$$\partial_{x_j} F_\alpha(x) = a_j \operatorname{Re}\int_{\mathbb{T}} e^{ij\theta} H_x(\theta)\,d\mu(\theta).$$

With the Fourier convention $\widehat{H}(k) = \int_{\mathbb{T}} e^{-ik\theta} H(\theta)\,d\mu(\theta)$, this is

$$\partial_{x_j} F_\alpha(x) = a_j \operatorname{Re}\widehat{H}_x(-j).$$

By Lemma 3.1, $a_j^2 \le C_a^2/n$. Therefore Parseval's identity gives

$$\sum_{j=0}^{n-1} |\partial_{x_j} F_\alpha(x)|^2 \le \frac{C_a^2}{n}\sum_{j\in\mathbb{Z}} |\widehat{H}_x(j)|^2 = \frac{C_a^2}{n}\int_{\mathbb{T}} |H_x(\theta)|^2\,d\mu(\theta).$$

Since

$$|H_x(\theta)|^2 = \frac{|Z_x(\theta)|^2}{(|Z_x(\theta)|^2+\alpha^2)^2} \le \frac{1}{|Z_x(\theta)|^2+\alpha^2},$$

the first estimate follows after taking $C_{\mathrm{grad}} \ge C_a^2$.

For the second estimate, differentiate once more:

$$\partial^2_{x_jx_j}F_\alpha(x) = \int_{\mathbb{T}} D^2\Phi(Z_x(\theta))\big(a_je^{ij\theta}, a_je^{ij\theta}\big)\, d\mu(\theta).$$

A direct computation gives an absolute Hessian bound

$$\big\|D^2\Phi(z)\big\|_{\mathrm{op}} \le \frac{C_\Phi}{|z|^2+\alpha^2}, \qquad C_\Phi := 3.$$

Thus

$$|\partial^2_{x_jx_j}F_\alpha(x)| \le C_\Phi a_j^2 \int_{\mathbb{T}} \frac{d\mu(\theta)}{|Z_x(\theta)|^2+\alpha^2}.$$

Taking $C_{\mathrm{grad}} \ge C_\Phi$ completes the proof. □

Lemma 6.2 (Discrete carré-du-champ)**.** *There are absolute constants $B_{\mathrm{flip}} \ge 1$ and $C_{\mathrm{cdg}} < \infty$ such that, if $B \ge B_{\mathrm{flip}}$, then for all sufficiently large n, uniformly for $r \in I_n$,*

$$\sum_{j=0}^{n-1} |\Delta_jF_\alpha|^2 \le C_{\mathrm{cdg}}\left(\frac{R_\alpha}{n} + \frac{R_\alpha^2}{n}\right).$$

Proof. Fix j. Along the coordinate segment from ε to $\tau_j\varepsilon$, the normalized polynomial changes by at most

$$|\Delta Z(\theta)| \le 2|a_j| \le C_\Delta n^{-1/2}, \qquad C_\Delta := 2C_a.$$

Set $B_{\mathrm{flip}} := 1$. Since $\alpha = (\log n)^B n^{-1/2}$ and $B \ge 1$, after increasing the threshold for n we have

$$C_\Delta n^{-1/2} \le \alpha/10.$$

If $|z-w| \le \alpha/10$, then

$$C_{\mathrm{path}}^{-1}(|w|^2+\alpha^2) \le |z|^2+\alpha^2 \le C_{\mathrm{path}}(|w|^2+\alpha^2).$$

Consequently,

$$\frac{1}{|Z_x(\theta)|^2+\alpha^2} \le C_{\mathrm{path}}\frac{1}{|Z_\varepsilon(\theta)|^2+\alpha^2}.$$

Consequently, for every point x on the coordinate segment from ε to $\tau_j\varepsilon$,

$$\int_{\mathbb{T}} \frac{d\mu(\theta)}{|Z_x(\theta)|^2+\alpha^2} \le C_{\mathrm{path}}R_\alpha(\varepsilon).$$

Taylor's formula in the j-th coordinate gives

$$\Delta_jF_\alpha = 2\varepsilon_j\partial_{x_j}F_\alpha(\varepsilon) + E_j,$$

where

$$|E_j| \le 2\sup_x |\partial^2_{x_jx_j}F_\alpha(x)|,$$

and the supremum is over the same coordinate segment. By Lemma 6.1,

$$|E_j| \le 2C_{\mathrm{grad}}C_{\mathrm{path}}a_j^2R_\alpha \le C_E\frac{R_\alpha}{n}, \qquad C_E := 2C_{\mathrm{grad}}C_{\mathrm{path}}C_a^2.$$

Using $|u+v|^2 \le 2|u|^2+2|v|^2$,

$$\sum_{j=0}^{n-1}|\Delta_jF_\alpha|^2 \le 8\sum_{j=0}^{n-1}|\partial_{x_j}F_\alpha(\varepsilon)|^2 + 2\sum_{j=0}^{n-1}|E_j|^2.$$

Applying Lemma 6.1 to the first term and the preceding bound to the second term gives

$$\sum_{j=0}^{n-1} |\Delta_j F_\alpha|^2 \le 8C_{\text{grad}} \frac{R_\alpha}{n} + 2C_E^2 \frac{R_\alpha^2}{n}.$$

The lemma follows with

$$C_{\text{cdg}} := 8C_{\text{grad}} + 2C_E^2.$$

□

Lemma 6.3 (Smoothed concentration)**.** *Assume $B \ge B_{\text{flip}}$. For this fixed B, there are constants $C_{\text{sm}}(B), \kappa_{\text{sm}}(B) < \infty$ such that, for all sufficiently large n and uniformly for $r \in I_n$,*

$$\boxed{\|F_\alpha - \mathbb{E}F_\alpha\|_{L^4(\Omega)} \le C_{\text{sm}}(B)(\log n)^{\kappa_{\text{sm}}(B)} n^{-1/2}.}$$

Proof. By Theorem 2.5, applied to $G = F_\alpha$,

$$\|F_\alpha - \mathbb{E}F_\alpha\|_{L^4(\Omega)}^2 \le C_{\text{Pis}} \left\| \sum_{j=0}^{n-1} |\Delta_j F_\alpha|^2 \right\|_{L^2(\Omega)}.$$

Using Lemma 6.2,

$$\|F_\alpha - \mathbb{E}F_\alpha\|_{L^4(\Omega)}^2 \le C_{\text{Pis}} C_{\text{cdg}} \left\| \frac{R_\alpha}{n} + \frac{R_\alpha^2}{n} \right\|_{L^2(\Omega)}.$$

Therefore

$$\|F_\alpha - \mathbb{E}F_\alpha\|_{L^4(\Omega)}^2 \le \frac{C_{\text{Pis}} C_{\text{cdg}}}{n} \left(\|R_\alpha\|_{L^2(\Omega)} + \|R_\alpha\|_{L^4(\Omega)}^2 \right).$$

By Proposition 5.1, this is at most

$$\frac{C_{\text{sm}}^2(B)(\log n)^{\kappa_{\text{sm}}^2(B)}}{n},$$

where one may take

$$C_{\text{sm}}^2(B) := C_{\text{Pis}} C_{\text{cdg}} \big(C_{\text{res}}(2, B) + C_{\text{res}}(4, B)^2 \big),$$

and

$$\kappa_{\text{sm}}^2(B) := \max\{\kappa_{\text{res}}(2, B), 2\kappa_{\text{res}}(4, B)\}.$$

Taking square roots proves the result with

$$C_{\text{sm}}(B) := \sqrt{C_{\text{sm}}^2(B)}, \qquad \kappa_{\text{sm}}(B) := \frac{1}{2}\kappa_{\text{sm}}^2(B).$$

□

7. Removing the smoothing

Write $M(u) = M_{n,r}(u)$ in this section. By Lemma 3.1, for all sufficiently large n,

$$\frac{1}{2}\Lambda \le A := \sigma_n(r)\alpha \le 2\Lambda.$$

Since P_n is not the zero polynomial, for each fixed $r > 0$ the set $\{\theta : f_r(\theta) = 0\}$ is finite, hence has μ-measure zero. Thus the following pointwise estimate may be applied for μ-a.e. θ, and the subsequent layer-cake identities are justified by Tonelli's theorem for nonnegative functions. For every $x > 0$,

$$\frac{1}{2} \log \left(1 + \frac{A^2}{x^2} \right) \le \frac{A^2}{x^2 + A^2} + \log_+ \frac{A}{x}.$$

Consequently,

$$0 \le F_\alpha - L_n \le \alpha^2 R_\alpha + T_A, \qquad T_A := \int_0^A M(u)\frac{du}{u}. \tag{7}$$

Lemma 7.1 (Truncation error)**.** *Assume $B \ge B_{\mathrm{flip}}$. There are constants $C_{\mathrm{tr}}(B), \kappa_{\mathrm{tr}}(B) < \infty$ such that, for all sufficiently large n and uniformly for $r \in I_n$,*

$$\boxed{\|F_\alpha - L_n\|_{L^4(\Omega)} \le C_{\mathrm{tr}}(B)(\log n)^{\kappa_{\mathrm{tr}}(B)} n^{-1/2}.}$$

Proof. We estimate the two terms in (7) separately.

First, by Proposition 5.1,

$$\left\|\alpha^2 R_\alpha\right\|_{L^4(\Omega)} = \alpha^2 \left\|R_\alpha\right\|_{L^4(\Omega)} \le C_{\mathrm{res}}(4,B)(\log n)^{\kappa_{\mathrm{res}}(4,B)+2B} n^{-1}.$$

Since $n^{-1} \le n^{-1/2}$, this is already of the required order.

Next split

$$T_A = T_{\mathrm{deep}} + T_{\mathrm{mid}}, \qquad T_{\mathrm{deep}} := \int_0^{n^{-1}} M(u)\frac{du}{u}, \qquad T_{\mathrm{mid}} := \int_{n^{-1}}^A M(u)\frac{du}{u}.$$

For the midrange term, Minkowski's integral inequality and Proposition 4.2 with $p = 4$ and $K_u = 1$ give

$$\|T_{\mathrm{mid}}\|_{L^4(\Omega)} \le C_{\mathrm{occ}}(4,1)\frac{1}{n}\int_{n^{-1}}^A \frac{(1+u)^2}{u}\,du,$$

because $A \le \sqrt{n}$ for all sufficiently large n. The integral is explicit:

$$\int_{n^{-1}}^A \frac{(1+u)^2}{u}\,du = \log(An) + 2(A - n^{-1}) + \frac{1}{2}(A^2 - n^{-2}).$$

Using $1 \le A \le 2\Lambda$ and $B \ge 1$, the last display is bounded by

$$C_{\mathrm{mid}}(B)(\log n)^{2B}, \qquad C_{\mathrm{mid}}(B) := 8,$$

for all sufficiently large n. Hence

$$\|T_{\mathrm{mid}}\|_{L^4(\Omega)} \le 8C_{\mathrm{occ}}(4,1)(\log n)^{2B} n^{-1}.$$

It remains to control the deep tail. Fubini's theorem gives

$$T_{\mathrm{deep}} = \int_{\{|f_r| \le n^{-1}\}} \log\frac{n^{-1}}{|f_r(\theta)|}\,d\mu(\theta) \le \int_{\{|f_r| \le n^{-1}\}} \log_+\frac{1}{|Z_r(\theta)|}\,d\mu(\theta),$$

where we used $f_r = \sigma_n(r) Z_r$ and $\sigma_n(r) \ge 1$. Define

$$B_{12} := \int_{\mathbb{T}} \log_+^{12}\frac{1}{|Z_r(\theta)|}\,d\mu(\theta).$$

Hölder's inequality in θ gives

$$T_{\mathrm{deep}} \le M(n^{-1})^{11/12} B_{12}^{1/12},$$

and therefore

$$T_{\mathrm{deep}}^4 \le M(n^{-1})^{11/3} B_{12}^{1/3}.$$

Hölder's inequality on Ω, with exponents 18/11 and 18/7, yields

$$\mathbb{E}T_{\mathrm{deep}}^4 \le (\mathbb{E}M(n^{-1})^6)^{11/18}(\mathbb{E}B_{12}^{6/7})^{7/18}.$$

By Jensen's inequality and Lemma 2.3,

$$(\mathbb{E}B_{12}^{6/7})^{7/18} \le (\mathbb{E}B_{12})^{1/3} \le C_{\log}(12)^{1/3}.$$

Also, Proposition 4.2 with $p = 6$ and $K_u = 1$ gives, for sufficiently large n,

$$\left\|M(n^{-1})\right\|_{L^6(\Omega)} \le 4C_{\mathrm{occ}}(6,1)n^{-1}.$$

Consequently,

$$\mathbb{E}T_{\mathrm{deep}}^4 \le \big(4C_{\mathrm{occ}}(6,1)\big)^{11/3} C_{\log}(12)^{1/3} n^{-11/3},$$

and hence

$$\|T_{\mathrm{deep}}\|_{L^4(\Omega)} \le \big(4C_{\mathrm{occ}}(6,1)\big)^{11/12} C_{\log}(12)^{1/12} n^{-11/12}.$$

Combining the three estimates in (7) proves the lemma. For instance, one may take

$$\kappa_{\mathrm{tr}}(B) := \max\{\kappa_{\mathrm{res}}(4,B) + 2B, 2B\},$$

and choose $C_{\mathrm{tr}}(B)$ larger than the sum of the three displayed constants above. □

8. Proof of KL4

Proof of Theorem 2.7. Choose once and for all

$$B_* := B_{\mathrm{flip}}.$$

The smoothed quantity F_α in this proof is formed with $B = B_*$. We have

$$L_n - \mathbb{E}L_n = (F_\alpha - \mathbb{E}F_\alpha) - ((F_\alpha - L_n) - \mathbb{E}(F_\alpha - L_n)).$$

Hence

$$\|L_n - \mathbb{E}L_n\|_{L^4(\Omega)} \le \|F_\alpha - \mathbb{E}F_\alpha\|_{L^4(\Omega)} + 2\,\|F_\alpha - L_n\|_{L^4(\Omega)}.$$

By Lemmas 6.3 and 7.1, the last display is bounded by

$$\big(C_{\mathrm{sm}}(B_*) + 2C_{\mathrm{tr}}(B_*)\big)(\log n)^{\max\{\kappa_{\mathrm{sm}}(B_*),\kappa_{\mathrm{tr}}(B_*)\}} n^{-1/2}.$$

Thus the theorem holds with

$$C_{\mathrm{KL}} := C_{\mathrm{sm}}(B_*) + 2C_{\mathrm{tr}}(B_*), \qquad \kappa_{\mathrm{KL}} := \max\{\kappa_{\mathrm{sm}}(B_*), \kappa_{\mathrm{tr}}(B_*)\}.$$

□

Proof of Corollary 2.8. By Theorems 2.4 and 2.7,

$$\|L_n(r) + \gamma/2\|_{L^4(\Omega)} \le \|L_n(r) - \mathbb{E}L_n(r)\|_{L^4(\Omega)} + \left|\mathbb{E}L_n(r) + \frac{\gamma}{2}\right|$$

$$\le \big(C_{\mathrm{KL}} + C_{\mathrm{Yakir}}\big)(\log n)^{\max\{\kappa_{\mathrm{KL}},2\}} n^{-1/2}.$$

Taking fourth powers proves the corollary with, for example,

$$C_{\mathrm{uKL}} := \big(C_{\mathrm{KL}} + C_{\mathrm{Yakir}}\big)^4, \qquad \kappa_{\mathrm{uKL}} := 4\max\{\kappa_{\mathrm{KL}}, 2\}.$$

□

9. Jensen's formula and the root count

Let

$$R_n^< = \#\{\zeta : P_n(\zeta) = 0,\ |\zeta| < 1\}, \qquad R_n^\le = \#\{\zeta : P_n(\zeta) = 0,\ |\zeta| \le 1\},$$

where zeros are counted with multiplicity. These quantities record the number of zeros of P_n inside, or inside and on, the unit circle.

For $r > 0$, define

$$J(r) = \int_{\mathbb{T}} \log |P_n(re^{i\theta})|\, d\mu(\theta),$$

the average of $\log |P_n|$ over the circle of radius r. This quantity measures the typical size of the polynomial on that circle.

By Jensen's formula, for any $r > 0$,

$$J(r) = \log |P_n(0)| + \sum_{|\zeta_j|<r} \log \frac{r}{|\zeta_j|},$$

where ζ_j are the zeros of P_n, counted with multiplicity. Each zero inside the disk contributes a term depending on its distance from the boundary. Since $|P_n(0)| = 1$, the constant term vanishes.

Lemma 9.1 (Deterministic normalizing term)**.** *There is an absolute constant $C_{\det} < \infty$ such that, for every $0 < \tau \le 1/2$ and every $n \ge 1$,*

$$\left| \frac{\log \sigma_n(1+\tau) - \log \sigma_n(1)}{\log(1+\tau)} - \frac{n}{2} \right| \le C_{\det}(1 + \tau n^2),$$

and

$$\left| \frac{\log \sigma_n(1) - \log \sigma_n(1-\tau)}{-\log(1-\tau)} - \frac{n}{2} \right| \le C_{\det}(1 + \tau n^2).$$

Proof. Put $s = \log r$ and

$$D(s) := \frac{d}{ds} \log \sigma_n(e^s) = \frac{\sum_{k=0}^{n-1} k e^{2ks}}{\sum_{k=0}^{n-1} e^{2ks}}.$$

Thus $D(s)$ is the average of $k \in \{0, \dots, n-1\}$ under the probability weights proportional to e^{2ks}. In particular,

$$D(0) = \frac{0 + 1 + \cdots + (n-1)}{n} = \frac{n-1}{2}.$$

Differentiating the weighted average gives

$$D'(s) = 2\,\mathrm{Var}_s(k),$$

where Var_s is the variance under the same weights. Since $0 \le k \le n-1$, we have $\mathrm{Var}_s(k) \le n^2$, and hence

$$|D'(s)| \le 2n^2 \qquad (s \in \mathbb{R}).$$

Let $h_+ = \log(1+\tau)$ and $h_- = -\log(1-\tau)$. For $0 < \tau \le 1/2$,

$$0 < h_+ \le \tau, \qquad 0 < h_- \le 2\tau.$$

By the fundamental theorem of calculus,

$$\frac{\log \sigma_n(1+\tau) - \log \sigma_n(1)}{\log(1+\tau)} = \frac{1}{h_+} \int_0^{h_+} D(s)\, ds,$$

and

$$\frac{\log \sigma_n(1) - \log \sigma_n(1-\tau)}{-\log(1-\tau)} = \frac{1}{h_-} \int_{-h_-}^{0} D(s)\, ds.$$

In both cases, the average differs from $D(0)$ by at most $2n^2 \max\{h_+, h_-\} \leq 4\tau n^2$. Since $|D(0) - n/2| = 1/2$, both desired estimates follow, for example with $C_{\mathrm{det}} := 5$. □

Lemma 9.2 (Jensen bounds near the unit circle)**.** *For* $0 < \tau < 1/2$*,*

$$\frac{J(1) - J(1-\tau)}{-\log(1-\tau)} \leq R_n^{<} \leq R_n^{\leq} \leq \frac{J(1+\tau) - J(1)}{\log(1+\tau)}.$$

Proof. We use Jensen's formula in the form

$$J(r) = \sum_{|\zeta_j|<r} \log \frac{r}{|\zeta_j|},$$

since $|P_n(0)| = 1$. The idea is to compare the increments of $J(r)$ as r moves slightly inward or outward from the unit circle, and interpret these increments in terms of contributions from zeros.

Upper bound on $R_n^{\leq}$. Consider the difference

$$J(1+\tau) - J(1).$$

By Jensen's formula, this equals

$$\sum_{|\zeta_j|<1+\tau} \log \frac{1+\tau}{|\zeta_j|} - \sum_{|\zeta_j|<1} \log \frac{1}{|\zeta_j|}.$$

Rewriting, we can group contributions zero-by-zero. Each zero contributes according to its modulus:

- If $|\zeta_j| \leq 1$, then the contribution is
$$\log \frac{1+\tau}{|\zeta_j|} - \log \frac{1}{|\zeta_j|} = \log(1+\tau).$$
Thus every zero in $|\zeta| \leq 1$ contributes exactly $\log(1+\tau)$.
- If $1 < |\zeta_j| < 1+\tau$, the zero does not appear in $J(1)$ but does appear in $J(1+\tau)$, contributing
$$\log \frac{1+\tau}{|\zeta_j|} \geq 0.$$
- If $|\zeta_j| \geq 1+\tau$, the zero contributes nothing to either term.

Therefore,

$$J(1+\tau) - J(1) \geq R_n^{\leq} \log(1+\tau),$$

since there are exactly $R_n^{\leq}$ zeros with $|\zeta_j| \leq 1$, each contributing $\log(1+\tau)$, and all other contributions are nonnegative. Dividing by $\log(1+\tau)$ gives

$$R_n^{\leq} \leq \frac{J(1+\tau) - J(1)}{\log(1+\tau)}.$$

Lower bound on $R_n^{<}$. Now consider

$$J(1) - J(1-\tau).$$

Again using Jensen's formula and comparing terms zero-by-zero, we classify contributions:

- If $|\zeta_j| < 1-\tau$, then the contribution is
$$\log \frac{1}{|\zeta_j|} - \log \frac{1-\tau}{|\zeta_j|} = -\log(1-\tau).$$
Thus each such zero contributes exactly $-\log(1-\tau)$.
- If $1-\tau \le |\zeta_j| < 1$, the zero appears in $J(1)$ but not in $J(1-\tau)$, contributing
$$\log \frac{1}{|\zeta_j|},$$
which lies between 0 and $-\log(1-\tau)$.
- If $|\zeta_j| \ge 1$, the zero contributes nothing to either term.

Thus every zero with $|\zeta_j| < 1$ contributes at most $-\log(1-\tau)$, and all other zeros contribute zero. Therefore,
$$J(1) - J(1-\tau) \le R_n^{<}\,[-\log(1-\tau)].$$
Dividing by $-\log(1-\tau)$ gives
$$\frac{J(1) - J(1-\tau)}{-\log(1-\tau)} \le R_n^{<}.$$

Middle inequality. The inequality
$$R_n^{<} \le R_n^{\le}$$
is immediate from the definitions, since the set $\{|\zeta| < 1\}$ is contained in $\{|\zeta| \le 1\}$.

Combining the three bounds completes the proof. □

Lemma 9.3 (Almost-sure control at the Jensen radii)**.** *Set*
$$\kappa_{\mathrm{root}} := \frac{\kappa_{\mathrm{uKL}} + 2}{8}, \qquad \tau_n := n^{-9/8}(\log n)^{\kappa_{\mathrm{root}}}, \qquad b_n := n^{-1/4}(\log n)^{2\kappa_{\mathrm{root}}}.$$
Then, with probability one,
$$\max_{r \in \{1-\tau_n,\, 1,\, 1+\tau_n\}} \left| L_n(r) + \frac{\gamma}{2} \right| \le b_n$$
for all sufficiently large n.

Proof. For all sufficiently large n,
$$\tau_n \le n^{-11/10},$$
since
$$\frac{\tau_n}{n^{-11/10}} = n^{-1/40}(\log n)^{\kappa_{\mathrm{root}}} \to 0.$$
Hence
$$1-\tau_n,\ 1,\ 1+\tau_n \in I_n$$
for all sufficiently large n.

By Corollary 2.8, uniformly over these three radii,
$$\mathbb{E}\left| L_n(r) + \frac{\gamma}{2} \right|^4 \le C_{\mathrm{uKL}}(\log n)^{\kappa_{\mathrm{uKL}}} n^{-2}.$$

By Markov's inequality and the union bound,
$$\mathbb{P}\left(\max_{r \in \{1-\tau_n, 1, 1+\tau_n\}} \left| L_n(r) + \frac{\gamma}{2} \right| > b_n \right) \le 3C_{\mathrm{uKL}}(\log n)^{\kappa_{\mathrm{uKL}}} n^{-2} b_n^{-4}.$$

Since
$$b_n^{-4} = n(\log n)^{-8\kappa_{\mathrm{root}}},$$

the right-hand side is

$$\leq 3C_{\mathrm{uKL}} n^{-1} (\log n)^{\kappa_{\mathrm{uKL}} - 8\kappa_{\mathrm{root}}}.$$

By definition of κ_{root},

$$\kappa_{\mathrm{uKL}} - 8\kappa_{\mathrm{root}} = -2,$$

so

$$\sum_{n=3}^{\infty} n^{-1} (\log n)^{-2} < \infty.$$

The Borel–Cantelli lemma completes the proof. □

Theorem 9.4 (Quantitative almost-sure root count)**.** *Set*

$$\kappa_{\mathrm{root}} := \frac{\kappa_{\mathrm{uKL}} + 2}{8}.$$

Then, with probability one, there exists a finite constant C_{as} such that

$$\left| R_n^{<} - \frac{n}{2} \right| + \left| R_n^{\leq} - \frac{n}{2} \right| \leq C_{\mathrm{as}} n^{7/8} (\log n)^{\kappa_{\mathrm{root}}}$$

for all sufficiently large n.

Proof. Set

$$\tau_n = n^{-9/8} (\log n)^{\kappa_{\mathrm{root}}}, \qquad b_n = n^{-1/4} (\log n)^{2\kappa_{\mathrm{root}}}.$$

Work on the probability-one event from the previous lemma.

Upper bound. By Lemma 9.2,

$$R_n^{\leq} \leq \frac{J(1+\tau_n) - J(1)}{\log(1+\tau_n)}.$$

Since $J(r) = \log \sigma_n(r) + L_n(r)$, Lemma 9.1 gives

$$R_n^{\leq} \leq \frac{n}{2} + C_{\mathrm{det}}(1 + \tau_n n^2) + \frac{L_n(1+\tau_n) - L_n(1)}{\log(1+\tau_n)}.$$

For large n,

$$\log(1+\tau_n) \geq \frac{\tau_n}{2},$$

hence

$$\left| \frac{L_n(1+\tau_n) - L_n(1)}{\log(1+\tau_n)} \right| \leq \frac{4b_n}{\tau_n}.$$

Lower bound. Similarly,

$$R_n^{<} \geq \frac{J(1) - J(1-\tau_n)}{-\log(1-\tau_n)}.$$

Thus

$$R_n^{<} \geq \frac{n}{2} - C_{\mathrm{det}}(1 + \tau_n n^2) - \frac{2b_n}{\tau_n}.$$

Since $R_n^{<} \leq R_n^{\leq}$, the upper bound for $R_n^{\leq}$ also gives an upper bound for $R_n^{<}$, while the lower bound for $R_n^{<}$ also gives a lower bound for $R_n^{\leq}$.

Conclusion. Since

$$\tau_n n^2 = n^{7/8} (\log n)^{\kappa_{\mathrm{root}}}, \qquad \frac{b_n}{\tau_n} = n^{7/8} (\log n)^{\kappa_{\mathrm{root}}},$$

we obtain

$$\left| R_n^{<} - \frac{n}{2} \right| + \left| R_n^{\leq} - \frac{n}{2} \right| \leq C_{\mathrm{as}} n^{7/8} (\log n)^{\kappa_{\mathrm{root}}}.$$

□

Corollary 9.5. *For every $\delta > 0$, with probability one there exists a finite constant C_δ such that*

$$\left|R_n^< - \frac{n}{2}\right| + \left|R_n^\leq - \frac{n}{2}\right| \leq C_\delta n^{7/8+\delta}$$

for all sufficiently large n.

Proof. Since $(\log n)^{\kappa_{\text{root}}} \leq n^\delta$ for large n, the result follows from the theorem. □

Corollary 9.6 (Erdős 522 formulation)**.** *Let*

$$Q_n(z) = \sum_{k=0}^{n} \varepsilon_k z^k,$$

and let $R_n(Q)$ be the number of zeros of Q_n in $\{|z| \leq 1\}$, counted with multiplicity. Then, for every $\delta > 0$, almost surely,

$$R_n(Q) = \frac{n}{2} + O_\delta(n^{7/8+\delta}).$$

In particular,

$$\frac{R_n(Q)}{n/2} \to 1$$

almost surely.

Proof. Apply the previous corollary to P_{n+1}. It gives

$$R_n(Q) = \frac{n+1}{2} + O_\delta(n^{7/8+\delta}) = \frac{n}{2} + O_\delta(n^{7/8+\delta}).$$

□

Use of AI Tools

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